c.) The manager of this convenience store tells you that the expected value of \$50 is not correct because It's impossible to withdraw \$50 from the ATM. How would you answer the manager?

~~Not a specific~~ Expected value is Long ~~average~~ run average withdraw amount not a specific withdraw and \$50 cannot be withdrawn.

Expect value is not a value that must actually occur. It's the long-run average of all withdrawls if many customers use the ATM

d) what is the probability that the amount withdrawn from the ATM will fall within one standard deviation (σ) of the mean?

$\mu = 50$

$SD = \sigma = 26.27$

$50 \pm 26.27 =$

$50 + 26.27 = 23.73$

$50 - 26.27 = 76.27$

x	F(x)
10	0.10
20	0.15
40	0.30
60	0.15
80	0.25
100	0.05

Values in Range that are close to mean 40 & 60

$P(23.73 \leq x \leq 76.27)$

Values between 23.73 and 76.27 are 40 & 6

Add their probability

$P(x = 40) + P(x = 60)$

$0.30 + 0.15 = 0.45$ or 45%

The probability that the withdraw amount falls within one SD of the mean is 0.45 (45%)